Though former US President Donald Trump withdrew from the agreement in 2017, the new US President Joe Biden re-entered the agreement soon after his inauguration and the global community is committed to roll back and modify processes to control global GHG emission to the extent necessary to limit the temperature rise to agreed levels.

Carbon Budget and Emissions Gap

CO_2 in the earth's atmosphere reached 409.8 parts per million (ppm) in 2019. From 287 ppm, it reached 296 ppm in 1900 and increased to 311 ppm in 1950. Thereafter, it rocketed thanks to heavy industrialization and urbanization all over the world, reaching 331.11 ppm in 1975 and 369.55 ppm in 2000. In the past nineteen years, the CO_2 level has increased by 40.3 ppm, an increase of over 2 ppm every year.

The pre-industrial level of CO_2 is approximately assumed to be 280 ppm, a little below the level of 287 ppm for 1850. While it is a complex science with many influencing factors, scientists broadly agree that doubling of pre-industrial CO_2 in the atmosphere will likely cause average global surface temperatures to rise between 1.5º and 4.5º Celsius. Doubling of the pre-industrial level of CO_2 of 280 ppm means CO_2 levels rising to 560 ppm.

Global temperature records are available since 1880. In line with a certain relationship between the rise in CO_2 levels in the atmosphere and rise in the earth's surface temperatures, an increase of 1.1º Celsius has been observed since 1880. As the rise in surface temperature leads to severe consequences, the Paris Agreement aims to keep global temperature increase well below 2º Celsius, hoping to limit it to 1.5º.

There is a certain relationship between the weight of carbon burnt and CO_2 emissions—in the process of burning, one atom of carbon combines